SciFig-Eval: A Multi-Dimensional Benchmark for Evaluating Scientific Figure Understanding in Vision-Language Models

Anonymous ACL submission

Abstract

We present SciFig-Eval, a benchmark for evaluating vision-language models (VLMs) on scientific figure understanding. Unlike existing benchmarks that focus on question answering, SciFig-Eval provides a multi-dimensional evaluation framework spanning description quality, visual robustness, and behavioural reliability. We evaluate 8 VLMs across 250 English scientific figures (bar charts, line plots, pie charts) using: (1) a checklist-based MQM evaluation with binding verification, validated against human judgement (Spearman $\rho = 1.0$); (2) robustness testing under 6 image transformations including in-paper page embedding; (3) caption bias resistance probes measuring susceptibility to misleading textual context; (4) psychology-informed hallucination probes (presupposition embedding, false premise anchoring, unanswerable domain questions); and (5) an Admittance-Resistance-Inductance (A-R-I) framework for selective blur experiments testing honesty under visual uncertainty. Our results reveal that description quality and behavioural reliability are largely independent dimensions — models ranking highest on MQM do not necessarily exhibit the strongest resistance to hallucination or caption bias. We find that GPT-5.2 and Gemini 3.1 Pro lead on resistance (0.81–0.91), while Phi-4-multimodal shows near-zero caption bias resistance (0.05), suggesting it relies almost entirely on textual context over visual evidence.

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